

Name: _____

A Level Mathematics : Mechanics

Topic 7: Kinematics

Date:

Time:

Total marks available:

Total marks achieved: _____

© 2025 Exam Papers Practice. All Rights Reserved

Questions

Q1.

At time t seconds, where $t \geq 0$, a particle P moves so that its acceleration $\mathbf{a} \text{ m s}^{-2}$ is given by

$$\mathbf{a} = 5t\mathbf{i} - 15t^{\frac{1}{2}}\mathbf{j}$$

When $t = 0$, the velocity of P is $20\mathbf{i} \text{ m s}^{-1}$

Find the speed of P when $t = 4$

(6)

(Total for question = 6 marks)

Q2.

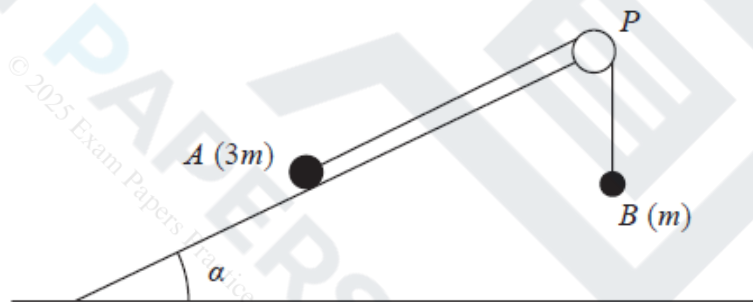


Figure 1

A small stone A of mass $3m$ is attached to one end of a string.

A small stone B of mass m is attached to the other end of the string.

Initially A is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

The string passes over a pulley P that is fixed at the top of the plane.

The part of the string from A to P is parallel to a line of greatest slope of the plane.

Stone B hangs freely below P , as shown in Figure 1.

The coefficient of friction between A and the plane is $\frac{1}{6}$

Stone A is released from rest and begins to move down the plane.

The stones are modelled as particles.

The pulley is modelled as being small and smooth.

The string is modelled as being light and inextensible.

Using the model for the motion of the system before B reaches the pulley,

(a) write down an equation of motion for A

(2)

(b) show that the acceleration of A is $\frac{1}{10}g$

(7)

(c) sketch a velocity-time graph for the motion of B , from the instant when A is released from rest to the instant just before B reaches the pulley, explaining your answer.

(2)

In reality, the string is not light.

(d) State how this would affect the working in part (b).

(1)

(Total for question = 12 marks)

Q3.

A particle P moves with constant acceleration $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$

At time $t = 0$, P is moving with velocity $4\mathbf{i} \text{ m s}^{-1}$

(a) Find the velocity of P at time $t = 2$ seconds.

(2)

At time $t = 0$, the position vector of P relative to a fixed origin O is $(\mathbf{i} + \mathbf{j}) \text{ m}$.

(b) Find the position vector of P relative to O at time $t = 3$ seconds.

(2)

(Total for question = 4 marks)

Q4.

(i) At time t seconds, where $t \geq 0$, a particle P moves so that its acceleration $\mathbf{a} \text{ m s}^{-2}$ is given by

$$\mathbf{a} = (1 - 4t)\mathbf{i} + (3 - t^2)\mathbf{j}$$

At the instant when $t = 0$, the velocity of P is $36\mathbf{i} \text{ m s}^{-1}$

(a) Find the velocity of P when $t = 4$

(3)

(b) Find the value of t at the instant when P is moving in a direction perpendicular to $\mathbf{i}$

(3)

(ii) At time t seconds, where $t \geq 0$, a particle Q moves so that its position vector $\mathbf{r}$ metres, relative to a fixed origin O , is given by

$$\mathbf{r} = (t^2 - t)\mathbf{i} + 3t\mathbf{j}$$

Find the value of t at the instant when the speed of Q is 5 m s^{-1}

(6)

(Total for question = 12 marks)

Q5.

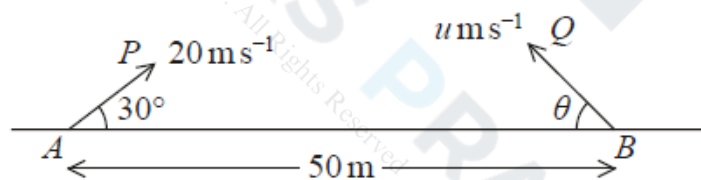


Figure 3

The points A and B lie 50 m apart on horizontal ground.

At time $t = 0$ two small balls, P and Q , are projected in the vertical plane containing AB .

Ball P is projected from A with speed 20 m s^{-1} at 30° to AB .

Ball Q is projected from B with speed $u \text{ m s}^{-1}$ at angle θ to BA , as shown in Figure 3.

At time $t = 2$ seconds, P and Q collide.

Until they collide, the balls are modelled as particles moving freely under gravity.

(a) Find the velocity of P at the instant before it collides with Q .

(6)

(b) Find

(i) the size of angle θ ,

(ii) the value of u .

(6)

(c) State one limitation of the model, other than air resistance, that could affect the accuracy of your answers.

(1)

(Total for question = 13 marks)

Q6.

Unless otherwise stated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

At time t seconds, where $t \geq 0$, a particle P moves in the x - y plane in such a way that its velocity $\mathbf{v} \text{ m s}^{-1}$ is given by

$$\mathbf{v} = t^{-\frac{1}{2}}\mathbf{i} - 4t\mathbf{j}$$

When $t = 1$, P is at the point A and when $t = 4$, P is at the point B .

Find the exact distance AB .

(6)

(Total for question = 6 marks)

Q7.

Unless otherwise stated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors due east and due north respectively and position vectors are given relative to the fixed point O .]

A particle P moves with constant acceleration.

At time $t = 0$, the particle is at O and is moving with velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$

At time $t = 2$ seconds, P is at the point A with position vector $(7\mathbf{i} - 10\mathbf{j}) \text{ m}$.

(a) Show that the magnitude of the acceleration of P is 2.5 m s^{-2}

(4)

At the instant when P leaves the point A , the acceleration of P changes so that P now moves with constant acceleration $(4\mathbf{i} + 8.8\mathbf{j}) \text{ m s}^{-2}$

At the instant when P reaches the point B , the direction of motion of P is north east.

(b) Find the time it takes for P to travel from A to B .

(4)

(Total for question = 8 marks)

Q8.

[In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 4 kg is at rest at the point A on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (4\mathbf{i} - \mathbf{j})\text{N}$ and $\mathbf{F}_2 = (\lambda\mathbf{i} + \mu\mathbf{j})\text{N}$, where λ and μ are constants, are applied to P

Given that P moves in the direction of the vector $(3\mathbf{i} + \mathbf{j})$

(a) show that

$$\lambda - 3\mu + 7 = 0$$

(4)

At time $t = 4$ seconds, P passes through the point B .

Given that $\lambda = 2$

(b) find the length of AB .

(5)

(Total for question = 9 marks)

Q9.

A particle, P , moves with constant acceleration $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$

At time $t = 0$, the particle is at the point A and is moving with velocity $(-\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$

At time $t = T$ seconds, P is moving in the direction of vector $(3\mathbf{i} - 4\mathbf{j})$

(a) Find the value of T .

(4)

At time $t = 4$ seconds, P is at the point B .

(b) Find the distance AB .

(4)

(Total for question = 8 marks)

Q10.

[In this question position vectors are given relative to a fixed origin O]

At time t seconds, where $t \geq 0$, a particle, P , moves so that its velocity $\mathbf{v} \text{ m s}^{-1}$ is given by

$$\mathbf{v} = 6t\mathbf{i} - 5t^{\frac{3}{2}}\mathbf{j}$$

When $t = 0$, the position vector of P is $(-20\mathbf{i} + 20\mathbf{j}) \text{ m}$.

(a) Find the acceleration of P when $t = 4$

(3)

(b) Find the position vector of P when $t = 4$

(3)

(Total for question = 6 marks)

Q11.

At time t seconds, a particle P has velocity $\mathbf{v}$ m s⁻¹, where

$$\mathbf{v} = 3t^{\frac{1}{2}}\mathbf{i} - 2t\mathbf{j} \quad t > 0$$

(a) Find the acceleration of P at time t seconds, where $t > 0$

(2)

(b) Find the value of t at the instant when P is moving in the direction of $\mathbf{i} - \mathbf{j}$

(3)

At time t seconds, where $t > 0$, the position vector of P , relative to a fixed origin O , is $\mathbf{r}$ metres.

When $t = 1$, $\mathbf{r} = -\mathbf{j}$

(c) Find an expression for $\mathbf{r}$ in terms of t .

(3)

(d) Find the exact distance of P from O at the instant when P is moving with speed 10 m s⁻¹

(6)

(Total for question = 14 marks)

Q12.

A particle P moves with acceleration $(4\mathbf{i} - 5\mathbf{j})$ m s⁻²

At time $t = 0$, P is moving with velocity $(-2\mathbf{i} + 2\mathbf{j})$ m s⁻¹

(a) Find the velocity of P at time $t = 2$ seconds.

(2)

At time $t = 0$, P passes through the origin O .

At time $t = T$ seconds, where $T > 0$, the particle P passes through the point A .

The position vector of A is $(\lambda\mathbf{i} - 4.5\mathbf{j})\text{m}$ relative to O , where λ is a constant.

(b) Find the value of T .

(4)

(c) Hence find the value of λ

(2)

(Total for question = 8 marks)

Q13.

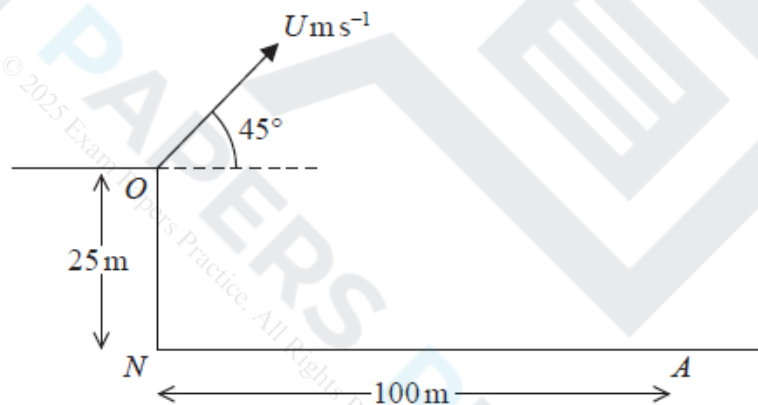


Figure 2

A small ball is projected with speed $U \text{ m s}^{-1}$ from a point O at the top of a vertical cliff.

The point O is 25 m vertically above the point N which is on horizontal ground.

The ball is projected at an angle of 45° above the horizontal.

The ball hits the ground at a point A , where $AN = 100 \text{ m}$, as shown in Figure 2.

The motion of the ball is modelled as that of a particle moving freely under gravity.

Using this initial model,

(a) show that $U = 28$

(6)

(b) find the greatest height of the ball above the horizontal ground NA .

(3)

In a refinement to the model of the motion of the ball from O to A , the effect of air resistance is included.

This refined model is used to find a new value of U .

(c) How would this new value of U compare with 28, the value given in part (a)?

(1)

(d) State one further refinement to the model that would make the model more realistic.

(1)

(Total for question = 11 marks)

Q14.

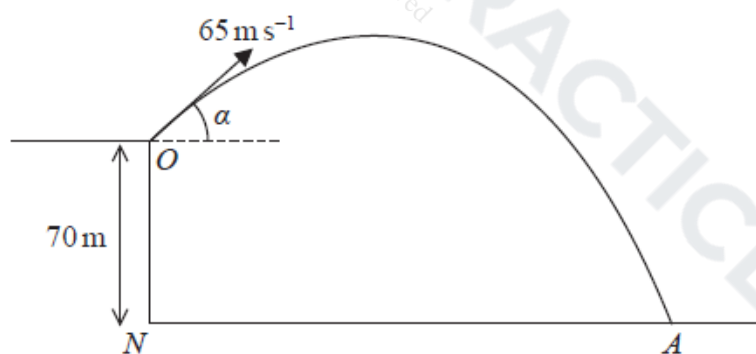


Figure 3

A small stone is projected with speed 65 m s^{-1} from a point O at the top of a vertical cliff.

Point O is 70 m vertically above the point N .

Point N is on horizontal ground.

The stone is projected at an angle α above the horizontal, where $\tan \alpha =$

The stone hits the ground at the point A , as shown in Figure 3.

The stone is modelled as a particle moving freely under gravity.

The acceleration due to gravity is modelled as having magnitude 10 m s^{-2}

Using the model,

(a) find the time taken for the stone to travel from O to A ,

(4)

(b) find the speed of the stone at the instant just before it hits the ground at A .

(5)

One limitation of the model is that it ignores air resistance.

(c) State one other limitation of the model that could affect the reliability of your answers.

(1)

(Total for question = 10 marks)

Q15.

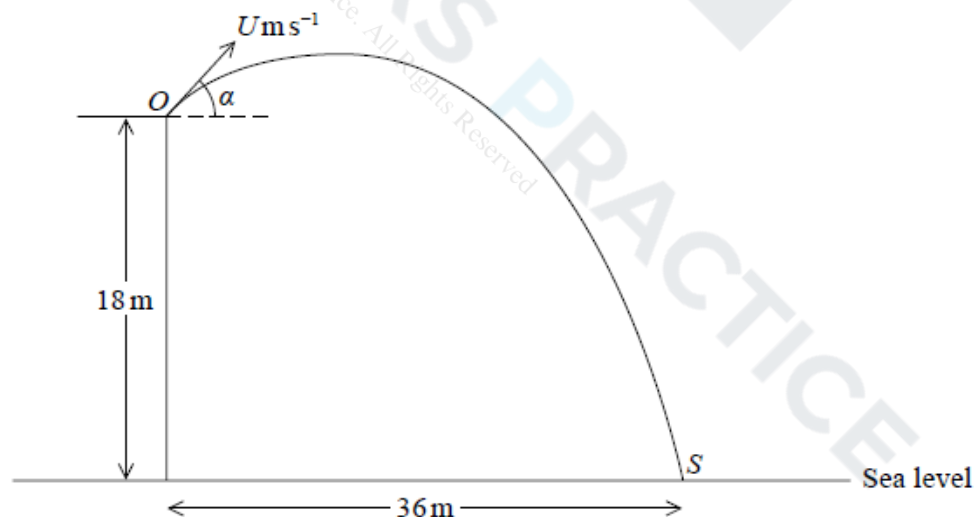


Figure 2

A boy throws a stone with speed $U \text{ m s}^{-1}$ from a point O at the top of a vertical cliff. The point O is 18 m above sea level.

The stone is thrown at an angle α above the horizontal, where $\tan \alpha = \frac{3}{4}$.

The stone hits the sea at the point S which is at a horizontal distance of 36 m from the foot of the cliff, as shown in Figure 2.

The stone is modelled as a particle moving freely under gravity with $g = 10 \text{ m s}^{-2}$

Find

(a) the value of U ,

(6)

(b) the speed of the stone when it is 10.8 m above sea level, giving your answer to 2 significant figures.

(5)

(c) Suggest two improvements that could be made to the model.

(2)

(Total for question = 13 marks)

Q16.

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors due east and due north respectively]

A radio controlled model boat is placed on the surface of a large pond.

The boat is modelled as a particle.

At time $t = 0$, the boat is at the fixed point O and is moving due north with speed 0.6 m s^{-1} .

Relative to O , the position vector of the boat at time t seconds is $\mathbf{r}$ metres.

At time $t = 15$, the velocity of the boat is $(10.5\mathbf{i} - 0.9\mathbf{j}) \text{ m s}^{-1}$.

The acceleration of the boat is constant.

(a) Show that the acceleration of the boat is $(0.7\mathbf{i} - 0.1\mathbf{j}) \text{ m s}^{-2}$.

(2)

(b) Find $\mathbf{r}$ in terms of t .

(2)

(c) Find the value of t when the boat is north-east of O .

(3)

(d) Find the value of t when the boat is moving in a north-east direction.

(3)

(Total for question = 10 marks)

Q17.

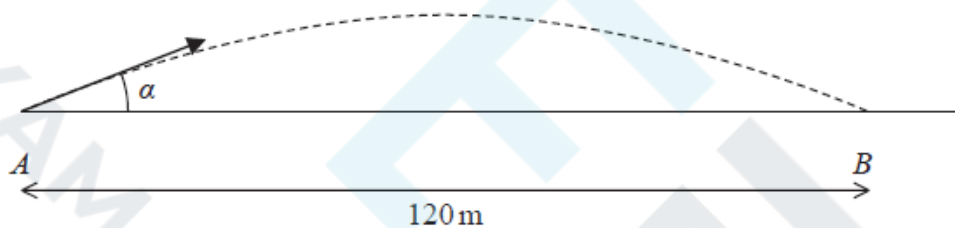


Figure 3

A golf ball is at rest at the point A on horizontal ground.

The ball is hit and initially moves at an angle α to the ground.

The ball first hits the ground at the point B , where $AB = 120$ m, as shown in Figure 3.

The motion of the ball is modelled as that of a particle, moving freely under gravity, whose initial speed is $U \text{ m s}^{-1}$

Using this model,

(a) show that $U^2 \sin \alpha \cos \alpha = 588$

(6)

The ball reaches a maximum height of 10 m above the ground.

(b) Show that $U^2 = 1960$

(4)

In a refinement to the model, the effect of air resistance is included.

The motion of the ball, from A to B , is now modelled as that of a particle whose initial speed is $V \text{ m s}^{-1}$

This refined model is used to calculate a value for V

(c) State which is greater, U or V , giving a reason for your answer.

(1)

(d) State one further refinement to the model that would make the model more realistic.

(1)

(Total for question = 12 marks)

Q18.

Unless otherwise stated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

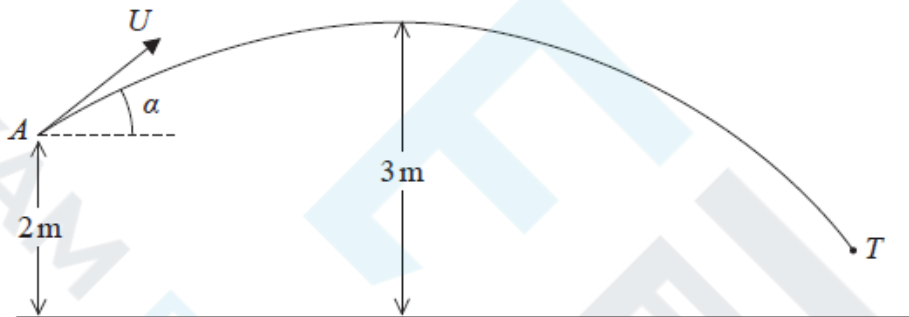


Figure 4

A boy throws a ball at a target. At the instant when the ball leaves the boy's hand at the point A , the ball is 2 m above horizontal ground and is moving with speed U at an angle α above the horizontal.

In the subsequent motion, the highest point reached by the ball is 3 m above the ground. The target is modelled as being the point T , as shown in Figure 4. The ball is modelled as a particle moving freely under gravity.

Using the model,

(a) show that $U^2 = \frac{2g}{\sin^2 \alpha}$.

(2)

The point T is at a horizontal distance of 20 m from A and is at a height of 0.75 m above the ground. The ball reaches T without hitting the ground.

(b) Find the size of the angle α

(9)

(c) State one limitation of the model that could affect your answer to part (b).

(1)

(d) Find the time taken for the ball to travel from A to T .

(3)

(Total for question = 15 marks)

Q19.

[In this question, position vectors are given relative to a fixed origin.]

At time t seconds, where $t > 0$, a particle P has velocity $\mathbf{v}$ m s⁻¹ where

$$\mathbf{v} = 3t^2\mathbf{i} - 6t^{\frac{1}{2}}\mathbf{j}$$

(a) Find the speed of P at time $t = 2$ seconds.

(2)

(b) Find an expression, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, for the acceleration of P at time t seconds, where $t > 0$

(2)

At time $t = 4$ seconds, the position vector of P is $(\mathbf{i} - 4\mathbf{j})$ m.

(c) Find the position vector of P at time $t = 1$ second.

(4)

(Total for question = 8 marks)